

```
dharanidhar@LAPTOP-2REPPIJ:~$ nano process_states.c
dharanidhar@LAPTOP-2REPPIJ:~$ cat process_states.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    printf("=== BEFORE FORK ===\n");
    printf("Parent PID   : %d\n", getpid());
    printf("Parent PPID  : %d\n", getppid());
    printf("State        : Parent is Running\n\n");

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }

    if (pid == 0)
    {
        printf("=== CHILD PROCESS ===\n");
        printf("Child PID   : %d\n", getpid());
        printf("Child PPID  : %d\n", getppid());
        printf("State        : Child is Running\n");

        printf("Child going to sleep...\n");
        sleep(10);

        printf("State        : Child is Running again\n");
        printf("Child terminating...\n");

        exit(0);
    }
    else

```

```

    }
else
{
    printf("=== PARENT PROCESS ===\n");
    printf("Parent PID : %d\n", getpid());
    printf("Child PID  : %d\n", pid);
    printf("State      : Parent is Waiting\n");

    wait(NULL);

    printf("State      : Child Terminated\n");
    printf("State      : Parent is Running again\n");
}

```

```

return 0;
}

```

```
dharanidhar@LAPTOP-2REPP...:~$ gcc process_states.c -o process_states
```

```
dharanidhar@LAPTOP-2REPP...:~$ ./process_states
```

```
=== BEFORE FORK ===
```

```
Parent PID : 2939
```

```
Parent PPID : 365
```

```
State      : Parent is Running
```

```
=== PARENT PROCESS ===
```

```
Parent PID : 2939
```

```
Child PID  : 2940
```

```
State      : Parent is Waiting
```

```
=== CHILD PROCESS ===
```

```
Child PID  : 2940
```

```
Child PPID : 2939
```

```
State      : Child is Running
```

```
Child going to sleep...
```

```
State      : Child is Running again
```

```
Child terminating...
```

```
State      : Child Terminated
```

```
State      : Parent is Running again
```

```
dharanidhar@LAPTOP-2REPP...:~$ ./process_states &
```

```
[1] 2950
```

```
dharanidhar@LAPTOP-2REPP...:~$ === BEFORE FORK ===
```

```
Parent PID : 2950
```

```
Parent PPID : 365
```

```

[1] 2950
dharanidhar@LAPTOP-2REPPLIJ:~$ === BEFORE FORK ===
Parent PID : 2950
Parent PPID : 365
State      : Parent is Running

=== PARENT PROCESS ===
=== CHILD PROCESS ===
Parent PID : 2950
Child PID  : 2951
State     : Parent is Waiting
Child PID : 2951
Child PPID : 2950
State     : Child is Running
Child going to sleep...
State     : Child is Running again
Child terminating...
State     : Child Terminated
State     : Parent is Running again

[1]+  Done                  ./process_states
dharanidhar@LAPTOP-2REPPLIJ:~$ ps -o pid,ppid,state,stat,cmd | grep process_states
  2978      365 S S+   grep --color=auto process_states
dharanidhar@LAPTOP-2REPPLIJ:~$ ps -f | grep process_states
dharani+  2987      365  0 07:39 pts/0    00:00:00 grep --color=auto process_states
dharanidhar@LAPTOP-2REPPLIJ:~$ grep -E "^(Name|State|Pid|PPid):" /proc/3001/status
grep: /proc/3001/status: No such file or directory
dharanidhar@LAPTOP-2REPPLIJ:~$ ./process_states & P=$!; sleep 1; C=$(pgrep -P $P); echo "Parent PID: $P"; echo "Child PID: $C";
d -p $P,$C; grep -E "^(Name|State|Pid|PPid):" /proc/$C/status
[1] 3006
=== BEFORE FORK ===
Parent PID : 3006
Parent PPID : 365
State      : Parent is Running

=== PARENT PROCESS ===
Parent PID : 3006
Child PID  : 3008
State     : Parent is Waiting
=== CHILD PROCESS ===
Child PID  : 3008

```

Child PID: 3008

```
PID      PPID S  STAT CMD
3006      365 S   S   ./process_states
3008      3006 S   S   ./process_states
```

Name: process_states

State: S (sleeping)

Pid: 3008

PPid: 3006

```

dharanidhar@LAPTOP-2REPPLIJ:~$ State      : Child is Running again

```

Child terminating...

```
State      : Child Terminated
```

```
State      : Parent is Running again
```

top

```
top - 07:46:44 up 1:33, 1 user, load average: 0.00, 0.00, 0.00
```

```
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
```

```
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
```

```
MiB Mem : 11803.7 total, 11082.1 free, 605.8 used, 289.9 buff/cache
```

```
MiB Swap: 3072.0 total, 3072.0 free, 0.0 used. 11197.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
3022	dharani+	20	0	9300	5768	3580	R	0.3	0.0	0:00.01
1	root	20	0	21952	13020	9324	S	0.0	0.1	0:01.82
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01
8	root	20	0	3228	2160	2008	S	0.0	0.0	0:00.61
176	root	20	0	4244	2688	2432	S	0.0	0.0	0:00.01
177	message+	20	0	9744	5612	4696	S	0.0	0.0	0:00.54
196	root	20	0	17980	8836	7780	S	0.0	0.1	0:00.22
198	root	20	0	1756104	13176	10520	S	0.0	0.1	0:00.26
213	syslog	20	0	222516	5960	4560	S	0.0	0.0	0:00.13
217	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00
241	root	20	0	107032	23300	13844	S	0.0	0.2	0:00.13
363	root	20	0	3188	1108	980	S	0.0	0.0	0:00.00
364	root	20	0	3204	1248	1108	S	0.0	0.0	0:00.10
365	dharani+	20	0	6596	5908	3660	S	0.0	0.0	0:00.13
366	root	20	0	6696	4648	3868	S	0.0	0.0	0:00.01
412	dharani+	20	0	20048	10168	8360	S	0.0	0.1	0:00.31
413	dharani+	20	0	21168	3592	1864	S	0.0	0.0	0:00.00
442	dharani+	20	0	6080	5360	3692	S	0.0	0.0	0:00.02
919	root	19	-1	50344	15384	14232	S	0.0	0.1	0:00.56
966	systemd+	20	0	91036	7596	6624	S	0.0	0.1	0:00.20
1019	root	20	0	24888	6268	5120	S	0.0	0.1	0:03.54

```
dharanidhar@LAPTOP-2REPPJ:~$ ps aux --sort=-rss
  919 root      19    -1    50344 15384 14232 S   0.0   0.1   0:00.56
  966 systemd+   20     0    91036  7596  6624 S   0.0   0.1   0:00.20
1019 root      20     0    24888  6268  5120 S   0.0   0.1   0:03.54
1069 systemd+   20     0    21344 12596 10468 S   0.0   0.1   0:00.10
1123 polkitd     20     0   308172  8100  7200 S   0.0   0.1   0:00.21
3023 root      20     0    24892  3392  2212 S   0.0   0.0   0:00.00
3024 root      20     0    24892  3392  2212 S   0.0   0.0   0:00.00
```

```
[1]+ Done ./process_states
dharanidhar@LAPTOP-2REPLIJ:~$ ps ...
error: garbage option
```

```
Usage:
ps [options]

Try 'ps --help <simple|list|output|threads|misc|all>'
or 'ps --help <s|l|o|t|m|a>'
for additional help text.
```

```
For more details see ps(1).
dharanidhar@LAPTOP-2REPLIJ:~$ ./process_states & P=$!; sleep 1; C=$(pgrep -P $P); ps -o pid,ppid,state,stat,cmd -p $
[1] 3034
=== BEFORE FORK ===
Parent PID : 3034
Parent PPID : 365
State      : Parent is Running

=== PARENT PROCESS ===
Parent PID : 3034
Child PID  : 3036
State      : Parent is Waiting
=== CHILD PROCESS ===
Child PID  : 3036
```

```
dharaanidhar@LAPTOP-2REPP: x + v

[1]+ Done ./process_states
dharaanidhar@LAPTOP-2REPLIJ:~$ ps ...
error: garbage option

Usage:
ps [options]

Try 'ps --help <simple|list|output|threads|misc|all>'
or 'ps --help <s|l|o|t|m|a>'
for additional help text.

For more details see ps(1).
dharaanidhar@LAPTOP-2REPLIJ:~$ ./process_states & P=$!; sleep 1; C=$(pgrep -P $P); ps -o pid,ppid,state,stat,cmd -p $P,$C
[1] 3034
=== BEFORE FORK ===
Parent PID : 3034
Parent PPID : 365
State : Parent is Running

=== PARENT PROCESS ===
Parent PID : 3034
Child PID : 3036
State : Parent is Waiting
=== CHILD PROCESS ===
Child PID : 3036
Child PPID : 3034
State : Child is Running
Child going to sleep...
  PID  PPID S  STAT CMD
  3034   365 S   S   ./process_states
  3036   3034 S   S   ./process_states
dharaanidhar@LAPTOP-2REPLIJ:~$ State : Child is Running again
Child terminating...
State : Child Terminated
State : Parent is Running again
dharaanidhar@LAPTOP-2REPLIJ:~$ |
```